Antoine Boucher

Software Engineer — Platform and graphics

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Summary

Software engineer focused on platform and graphics. Day-to-day work spans Kubernetes, GCP, Terraform, GitLab CI/CD, and observability for Java/Kotlin/Spring Boot and Python microservices, with JUnit, Playwright, and DevSecOps on releases. Master's at ÉTS: real-time rendering, physics simulation, and tactile sensing in C++, OpenGL/GLSL, and Blender.

Competences

Languages	Java, Kotlin, Python, C++, JavaScript
Backend & APIs	Spring Boot, microservices · Int.: C#, .NET, Android, GraphQL, Django
Graphics / 3D	OpenGL, GLSL, C++, real-time rendering · Int.: Blender, CUDA, OpenCV
Web	Angular, React, TypeScript
Cloud & platform	Docker, Kubernetes, Terraform, GCP, Azure, AWS
Delivery	Git, GitLab CI/CD, GitHub Actions · Int.: Azure DevOps
Data	PostgreSQL, MongoDB, Redis, Kafka, SQL
Reliability	Playwright, JUnit, ELK, Grafana, Prometheus
Security & practice	Linux, OWASP, Agile/SCRUM/Kanban, Open-source maintainer (uml-mcp)
Certifications	AWS Certified Cloud Practitioner (to Apr 2027); QcES; Cloud Quest: Cloud Practitioner, Data Analytics, Machine Learning, Serverless Developer, Networking, Solutions Architect

Experience

La Caisse (Caisse de dépôt et placement du Québec) *Montréal, QC*

Software Developer Intern — Digital and Analytics Solutions *Sep 2026 – Dec 2026*

IMC2 – Institut multidisciplinaire en cybersécurité et cyberrésilience *Montréal, QC (Remote)*

Secure cloud platforms *Mar 2025 – Feb 2026*

- Split a monolith into per-service repos and ran 5 GitLab CI/CD pipelines with dedicated runners on two GKE clusters in GCP; Terraform handled networking and observability for repeatable cloud releases
- Containerized workloads on Docker, ran research GKE on GCP (workload networking), and rolled out services through Terraform to ship infrastructure changes with less risk
- Added JUnit and Playwright E2E coverage to 5 Java microservices; wired Android builds through GitLab into the Terraform-managed GKE stack so mobile and backend releases stayed aligned
- Built scripted cybersecurity risk analysis across 1000+ risks and 181+ threat categories (weighted scoring, star-based prioritization, Angular triage UI) so analysts triaged from consistent context
- **Technical Skills:** Java, Kotlin, Android, Docker, Kubernetes, Terraform, GitLab CI/CD, GCP, JUnit, Playwright, Angular
- **Soft Skills:** Cross-functional collaboration, stakeholder communication, prioritization with research analysts

Polytechnique Montréal *Montréal, QC (On-site)*

Teaching Assistant — DevSecOps (LOG 8100) *Sep 2024 – Jan 2026*

- Built and maintained the GitLab workspace (repos, CI/CD templates, runners, workflows) for approximately 20 concurrent teams, cutting admin overhead during the term
- Coached Docker and Kubernetes build, deploy, and debug under LOG 8100; used OWASP-driven examples tied to hands-on lab scenarios
- Refreshed DevSecOps-aligned lab artifacts, graded practicals, and coached resubmissions for faster, cleaner turnaround
- **Technical Skills:** GitLab, CI/CD, Docker, Kubernetes, OWASP, Linux
- **Soft Skills:** Team coaching, constructive feedback, classroom facilitation

École de technologie supérieure (ÉTS) *Montréal, QC (Hybrid)*

Teaching Assistant *Feb 2024 – Apr 2026*

- Delivered labs, integrator-project support, and office hours on mobile/UX, integrator-project, and distributed-data tracks (e.g., TCH 057, IND 500, TCH 099, GTI 660, GTI 320)
- Authored and refreshed SQL, MongoDB, Kafka, and PostgreSQL exercises for IND 500, GTI 660, and related cohorts, with rubrics aligned to course outcomes
- Unblocked integrator teams on architecture and backend/data; kept Azure-backed course infrastructure running; SCRUM coaching on mobile/UX and GTI 320 so students could ship end-to-end features
- **Technical Skills:** SQL, MongoDB, Kafka, PostgreSQL, Azure, Android Studio, SCRUM
- **Soft Skills:** Student mentorship, rubric-driven feedback, patience under deadline pressure

IONODES *Laval, QC (Hybrid)*

Cloud Developer *May 2023 – Aug 2023*

- Designed a three-tier Auth0 subscription model for organization-scoped IoT access, gating product capabilities per tenant
- Integrated Sentry in C# with Serilog for structured production errors, making production issues easier to trace for support and engineering
- Advanced ONVIF and WebRTC streaming at IoT scale; ran Azure DevOps with feature flags and repeatable promotions for safer releases
- **Technical Skills:** C#, .NET, Azure DevOps, Auth0, Sentry, ONVIF, WebRTC
- **Soft Skills:** Requirements clarification, partner communication, calm under production pressure

Intact *Montréal, QC (Hybrid)*

Back End Developer (Co-op) *Jan 2022 – May 2022*

- Delivered a 6-step signup flow and mobile-app observability on Kotlin/Spring Boot microservices serving 10,000+ active users, improving activation for new accounts
- Implemented Kotlin login and MFA APIs for client mobile apps while preserving policy controls
- Extended the ELK stack with Kibana dashboards and harmonized microservice error codes so operations could triage incidents faster
- **Technical Skills:** Kotlin, Spring Boot, GraphQL, MongoDB, ELK, Kibana
- **Soft Skills:** Agile teamwork, incident communication, cross-team coordination

Power Go *Montréal, QC (Remote)*

Full Stack Web Developer (Co-op) *Sep 2020 – Apr 2021*

- Built a Python/Django/Vue.js/PostgreSQL feed parser unifying XML/JSON/CSV/Excel supplier feeds, integrated Facebook Shop and Marketplace APIs, and shipped a CAD/USD FX-rate service tied to inventory dates so cross-border catalogs stayed in sync without double entry
- **Technical Skills:** Python, Django, Vue.js, PostgreSQL, Git
- **Soft Skills:** Remote collaboration, supplier-facing problem solving

Wandrian *Montréal, QC (On-site)*

Full Stack Developer (Co-op → Part-time) *May 2019 – Jan 2020*

- On the Italiarail stack with Railkey Tech, integrated Braintree and extended purchase APIs while rebuilding the Selenium E2E harness and Django/Pyramid/jQuery internal tools for steadier bookings and earlier UI regression catches
- **Technical Skills:** Python, Django, Selenium, MongoDB, Git
- **Soft Skills:** QA partnership, detail-oriented delivery

Algolux *Montréal, QC (On-site)*

Engineering Intern *May 2017 – Aug 2017; May 2018 – Aug 2018*

- Built C++/CUDA/OpenCV imaging tooling for camera/ISP validation (HDR Nexus 6 rigs, Python RAW-color processing, Flask REST capture) and produced an approximately 12,000-image outdoor vision dataset, cutting manual validation time
- **Technical Skills:** Python, C++, CUDA, OpenCV, Flask, Linux
- **Soft Skills:** Research discipline, clear lab documentation

Education

École de technologie supérieure (ÉTS) *Montréal, QC, Canada*

Master's Degree, Information Technology Engineering *Aug 2023 – Aug 2026*

- Thesis: real-time surface wear in interactive physics simulations, with dynamic friction and textures.

École de technologie supérieure (ÉTS) *Montréal, QC, Canada*

Bachelor's Degree, Information Technology Engineering *Sep 2018 – Apr 2023*

- AlgoÉTS (ÉTS algorithmic trading club): President (Jan 2023 – Jan 2025); club infrastructure, web, and trading/data projects.
- **Academic focus:** software engineering, distributed systems, databases, and team-based integrator projects in ÉTS labs and coursework.